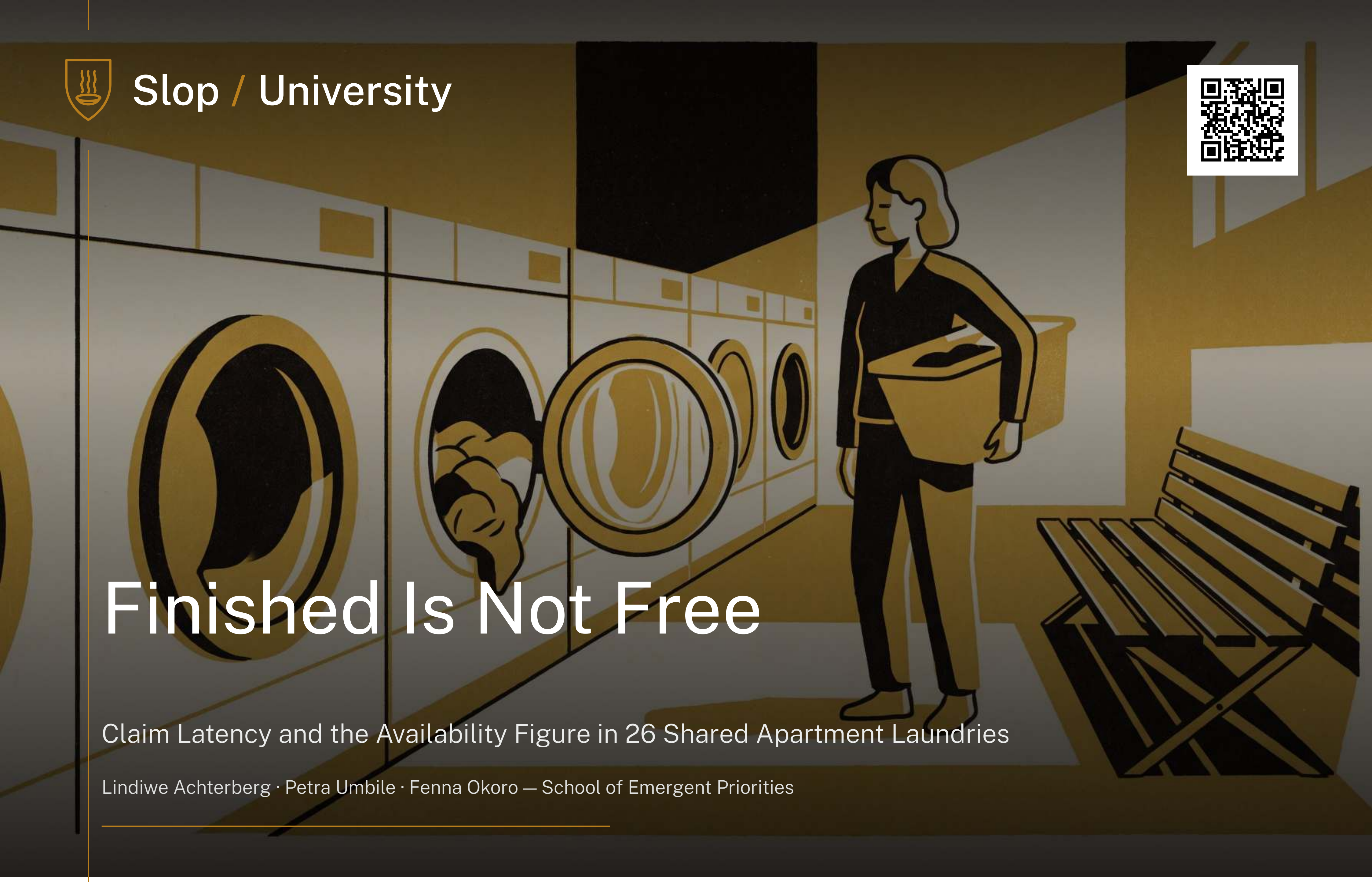

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Finished Is Not Free

Claim Latency and the Availability Figure in 26 Shared Apartment Laundries

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Motivation

A shared laundry is run from a booking platform that calls a machine available whenever no cycle is turning. A finished load nobody has collected reads exactly like an empty drum. How wide is the gap, and which side of it does the committee decide from?

Research questions

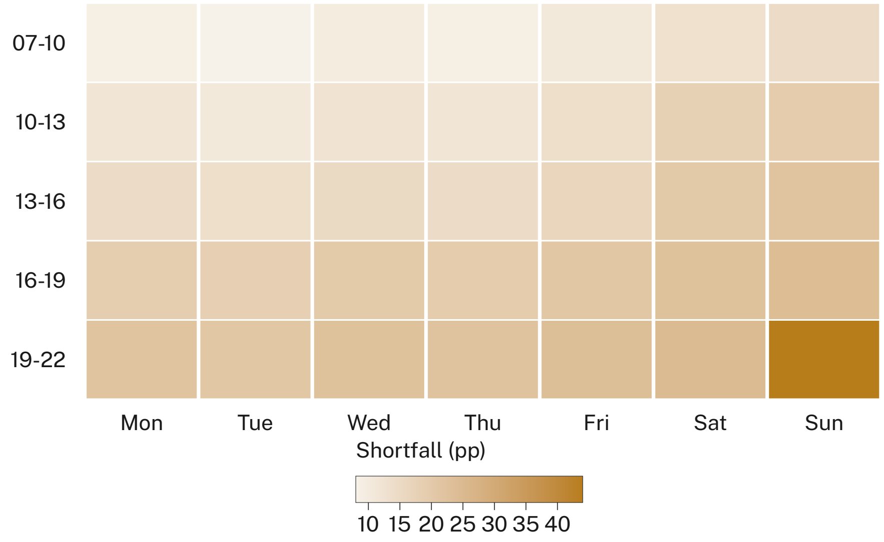
- How long do completed loads sit before collection?
- How far does effective availability fall below the reported figure?
- Does the shortfall concentrate in the hours capacity decisions rest on?

Study design

- 26 shared laundries · 78 washers · door and cycle-end loggers · 14 weeks · 38,441 completed cycles
- Discrete-event queue model calibrated on observed claim latency; effective availability counts a machine only when it is both idle and empty
- Availability windows and the two-hour unclaimed cut-off pre-registered; vendor logs ingested unaltered, none adjusted post hoc

What we found

Machines finish long before they empty. Median claim latency is 34 minutes (IQR 11–96), and 22.6% of loads are still in the drum at two hours. Reported availability averages 71.4% against an effective 48.2%.



Availability shortfall — reported minus effective — across 26 laundries over 14 weeks. The gap widens through the day and through the week, peaking at 44 percentage points on Sunday between 19:00 and 22:00, where the platform reports 63% availability against an effective 19%.

Implications & takeaways

Availability is measured on the wrong state: idle rather than usable. The shortfall is widest exactly where capacity is contested, so a committee reading the platform sees slack that residents queue through. No machine was added anywhere in the portfolio over the study period.

The gap is not evenly distributed. Effective availability in the four laundries with the longest median latency sits below half the portfolio figure, yet each reports the same headline number upward. The measure travels well between buildings; the wait does not.

Next: whether a collection prompt moves claim latency, or only the volume of complaints.

Selected references

1. Larson, R. C. (1987). Perspectives on queues: social justice and the psychology of queueing. *Operations Research*, 35(6), 895–905. doi:10.1287/opre.35.6.895
2. Little, J. D. C. (1961). A proof for the queuing formula: L = λW. *Operations Research*, 9(3), 383–387. doi:10.1287/opre.9.3.383
3. Kim, S.-H., & Whitt, W. (2014). Are call center and hospital arrivals well modeled by nonhomogeneous Poisson processes? *Manufacturing & Service Operations Management*, 16(3), 464–480. doi:10.1287/msom.2014.0490
4. Ostrom, E. (2010). Beyond markets and states: polycentric governance of complex economic systems. *American Economic Review*, 100(3), 641–672. doi:10.1257/aer.100.3.641
5. Melfi, R., Rosenblum, B., Nordman, B., & Christensen, K. (2011). Measuring building occupancy using existing network infrastructure. *2011 International Green Computing Conference*. doi:10.1109/IGCC.2011.6008560
6. Solberg, T., Nwosu, J., & Chikere, I. (2026). Still on the Walk Sheet: a controlled interrupted time-series study of orphaned quarterly meter reads after automated address normalisation. *School of Continuous Improvement, Slop University*. doi:10.5555/slop.04vwr3

We thank the residents of the instrumented laundries and the managing agents who granted access.

De-identified cycle telemetry only; no garment imaging and no resident identifiers retained. Latency windows pre-registered.

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